

SHUNYA

Gaming Engine

Architecture, Funnel Design & Game Scope Document

Overview & Architecture

At the core of Shunya lies its proprietary gaming engine, engineered to function as a high-precision elimination funnel. The platform begins with millions of registered participants and employs a multi-stage, skill-based filtration system that culminates in a single, ultimate winner.

The engine is built on a philosophy of progressive difficulty — each stage evaluates a deeper cognitive dimension, ensuring that the surviving participants are not merely lucky, but genuinely the most capable players on the platform.

Funnel Architecture at a Glance

Phase	Stages	Format	Target Outcome
Pre-Bucketing	Stages 1–3	Online (weekly cohorts)	Eliminate ~87.5% of participants per cohort
Post-Bucketing	Stages 4–5	Online (merged pool)	Whittle down to a few hundred finalists
Finale	Offline Final	Physical / In-Person	Crown the single ultimate winner

The Online Phase

The online phase manages the bulk of all eliminations. It is designed to handle massive concurrent user loads while maintaining competitive integrity. The phase is split into two sub-categories based on how participants are grouped — Pre-Bucketing and Post-Bucketing.

A. Pre-Bucketing (Stages 1 to 3)

To ensure fair matchmaking and manageable server loads during the initial surge, participants are grouped into cohorts (called "Buckets") based on the calendar week in which they register.

- **Weekly Cohorts:** Upon app launch, users are capped and grouped by registration week. All users who register in Week 1 form the "Week 1 Bucket," Week 2 users the "Week 2 Bucket," and so on. New registrations close completely after Week 5 or Week 6 (configurable).
- **Stage Progression:** Within their specific weekly window, each bucket must complete a defined set of games per stage.
- **The 50% Elimination Rule:** At the end of each of the three pre-bucketing stages, approximately the bottom 50% of performers are eliminated. The elimination percentage is configurable by operators.

Example Scenario: If 1,000,000 users register in Week 1: End of Stage 1 → 500,000 remain. End of Stage 2 → 250,000 remain. End of Stage 3 → 125,000 advance.

B. Post-Bucketing (Stages 4 to 5)

Once all weekly cohorts have completed the three pre-bucketing stages, the surviving participants are de-siloed from their registration cohorts.

- **Merging the Cohorts:** All remaining participants from Weeks 1 through 6 are merged into a single, unified competitive pool.
- **Final Online Stages:** This merged pool competes head-to-head across two final online stages (Stages 4 and 5), evaluated on advanced metrics like persistence, strategy, and speed.
- **Outcome:** By the end of Stage 5, the online engine has successfully narrowed the field from millions to a few hundred elite qualifiers.

The Offline Phase (The Finale)

The final few hundred participants who survive all five online stages are personally invited to a physical venue to compete in the ultimate showdown.

- **Physical Competition:** Elite players face off in person across a curated set of offline games designed for a live audience and a high-pressure environment.
- **The Ultimate Winner:** This physical stage is the final tier of the Shunya funnel — the single ultimate champion is crowned here.

STAGE 1 | "The Divine Echo of Precision" प्रथम

चरण | अचकता की दव्य गजं

"Your memory is your only weapon. The vault remembers everything — do you?"

Stage 1 tests a player's raw memory and knowledge retention. Drawing inspiration from classic memory games such as Simon Says and Memory Matching, and quiz formats used in competitive trivia platforms (e.g., Kahoot!, Quizlet Live), this stage rewards accuracy over speed. Players who advance are those who can faithfully recall and reconstruct information under pressure.

Stage 1 Game Summary Table

#	Game	Time Window	Evaluation Criteria	Shunya Points
1	Object Placement Memory (4x1)	Display: 5 secs Play: 30 secs (max)	Win / Lose	1 correct placement = 20 points Total points = n*m (80)
2	Sequence Recall	3 mins (max)	Win / Lose	100
3	Spot the Difference	3 mins (max)	Correct Finds + Speed	Up to 100
4	MCQ — General Knowledge	2 mins (time-bound)	Correct Answers	Easy: 5 Med: 10 Hard: 20

Stage Qualification: Only the top ~50% of scorers (by cumulative Shunya Points) advance to Stage 2. The percentage is operator configurable.

Game 1: Object Placement Memory

Inspired by: Classic grid-based memory games such as "Concentration" and spatial recall challenges used in neuropsychological assessments.

Objective: Memorize the positions of $n \times m$ objects on a $n \times m$ grid, then accurately place them back after they disappear.

- The screen displays a $n \times m$ grid ($n \times m$ cells total).
- Four distinct, randomized objects (e.g., fruits, animals, symbols) are placed — one in each cell — and shown to the player for exactly 't1' seconds.
- After 't1' seconds, all objects are removed from the grid and placed in a randomized tray at the bottom of the screen along with some other objects which were not part of the grid.
- The player must drag and drop each object back into its original cell.
- Result:
 - o 1 correct placement = 20 points
 - o Total points = $20 * n * m$
- Difficulty can be scaled in later stages by increasing grid size (e.g., 3x3, 4x4) or reducing the display duration.
- Objects displayed should be day to day objects so that they can be recalled easily.

Design Note: Object sets should be diverse and culturally neutral. Each game session should use a freshly randomized set to prevent pattern memorization across replays.

Reference: memozor.com — Grid Picture Memory Game

Game 2: Sequence Recall

Inspired by: The classic "Simon" electronic memory game (Hasbro, 1978) and the Corsi Block-Tapping Test used in cognitive psychology research.

Objective: Observe a sequence of 4 colored balls flashing in order, then reconstruct that exact sequence from memory.

- **4-cell horizontal grid** — one row containing four colored cells: Green, Red, Blue, and Yellow.
- **Unique sound per cell** — each cell plays a distinct tone when activated, whether by the computer or the player. (Very important as sound will be used to memorize the sequence)
- **Computer flashes one new cell per round** — each round; the computer picks up a random cell and flashes it briefly. It does NOT replay previous cells.
- **Player repeats the full sequence** — after each flash, the player must click all cells in the correct order from round 1 up to the current round.
- **Sequence grows one each round** — if the player succeeds, a new random cell is added and flashed; the player then repeats the longer sequence.

- **Wrong input ends the game** — if the player clicks the wrong cell at any point, the game immediately ends and displays the final score.
- **Visual feedback on activation** — cells light up with a glow effect when flashed by the computer or clicked by the player.
- Scoring:
 - o Length of the sequence * 20 (For example if the length of the sequence is 5 cells then resulting score is 5*20 i.e 100)

Reference: [Simon Says — Classic Sequence Memory \(mathsisfun.com\)](https://mathsisfun.com/games/simonsays.html)

The above link is just for reference. The requirements mentioned supersede.

Game 3: Spot the Difference

Inspired by: Visual discrimination tasks used in IQ assessments and classic "spot the difference" puzzle formats popularized in print and digitally (e.g., CrazyGames "What's the Difference").

Objective: Identify all predefined differences between two near-identical images as quickly as possible, within a 3-minute window.

- Two near-identical images are displayed side-by-side on screen, with a fixed number of predefined difference regions embedded in one image.
- The player taps the region on either image where they believe a difference exists.
- Correct tap: the difference is highlighted and locked; points are awarded. Incorrect tap: a penalty is applied and the miss is recorded.
- Scoring system (capped at 100 pts per game):
 - o Base points per correct find (e.g., 10–15 pts depending on difficulty).
 - o Speed bonus for completing all differences ahead of schedule.
 - o Penalty deduction for each incorrect tap.
 - o Hint usage deducts a fixed number of points per hint used.
- Game state tracking: number of differences found, remaining differences, time elapsed, and tap attempts are all tracked throughout the session.
- Difficulty scaling: Increase the number of differences, reduce their visual prominence, or decrease the available time window in later stages.

Reference: [CrazyGames — What's the Difference Online](https://crazygames.com/games/spot-the-difference)

Game 4: MCQ — General Knowledge

Inspired by: Adaptive quiz formats used by platforms like Duolingo (XP-based scoring), Quizlet Live, and Britannica's skill-based trivia systems.

Objective: Answer as many correctly-tiered multiple-choice general knowledge questions as possible within a 2-minute time window.

- Questions are drawn from a pre-curated General Knowledge bank spanning history, science, geography, culture, and current affairs.
- The difficulty system operates on three tiers: Easy (5 pts), Medium (10 pts), Hard (20 pts).
- Adaptive Difficulty Engine:
 - 2 consecutive correct answers → level increases (Easy → Medium → Hard).
 - 2 consecutive incorrect answers → level decreases (Hard → Medium → Easy).
- Both questions and answer options can contain images (e.g., identify a flag, a landmark, or a famous personality).
- Score = sum of points from all correct answers within the 2-minute window.
- No negative marking in Stage 1.

STAGE 2 | "The Kinetic Pulse"

द्वितीय चरण | रफ्तार का महाराज

"Thought is a luxury. Here, only your instincts survive."

Stage 2 shifts the evaluation lens from pure memory to reaction time and spatial problem-solving speed. Drawing from competitive puzzle gaming (e.g., Tetris, 2048, Rush Hour), this stage rewards players who think fast and execute accurately. Hesitation is penalised — the faster a player solves a challenge, the higher their score.

Stage 2 Game Summary Table

#	Game	Time Window	Evaluation Criteria	Shunya Points
1	Sliding Puzzle (3x3 with numbered tiles)	5 mins (max)	Reaction Time	0 if solved ≤5s; else (Window – Time Taken)
2	Colour-Based Sorting	3 mins (max)	Completion + Efficiency	Up to 100 (base + speed + efficiency bonuses)

3	Block Fill	3 mins (max)	Completion + Speed	Time Window – Time Taken
4	MCQ — Mathematics	2 mins (time-bound)	Correct Answers	Easy: 5 Med: 10 Hard: 20

Scoring Note (Reaction Time Games): Score = (Time Window in seconds) – (Time Taken in seconds). If solved in 0–5 seconds, score = 0 (anti-exploit guard). Example: Solved in 47s on a 120s window → Score = 73 pts.

Game 1: Sliding Puzzle

Inspired by: The classic 15-Puzzle (invented 1878), popularised digitally on early Nokia phones and modern apps like "Slide Puzzle — Classic Game" on iOS/Android.

Objective: Reassemble a scrambled image on a 3x3 grid by sliding tiles into the correct positions as fast as possible.

- The screen displays a 3x3 grid containing 9 tiles (one tile is empty to allow movement).
- A completed reference image is flashed to the player for 5 seconds to memorise the target state.
- After 5 seconds, the tiles are algorithmically scrambled. Solvability is always guaranteed via parity check.
- The player slides tiles into the empty space to reconstruct the original image.
- A move counter and timer are displayed; only the timer affects scoring.
- Result: Successful reconstruction = Win. Score = Time Window (120s) minus Time Taken. Unsolved within 2 minutes = Lose (0 pts).
- Image bank: Use abstract patterns rather than real images to avoid recognition bias.

Reference: [Sliding Puzzle — Brain Training \(helpfulgames.com\)](https://helpfulgames.com/sliding-puzzle/)

Game 2: Colour-Based Sorting

Inspired by: "Ball Sort Puzzle" (AppStore Top Chart, 2020–present) and colour-classification cognitive tasks used in attention research.

Objective: Sort all coloured balls into their matching test tubes — each tube must contain only one colour when complete.

- The screen presents multiple test tubes (stack structures), each partially filled with coloured balls arranged in a mixed order.
- Only the top ball of any tube can be moved. A ball may only be placed on another tube if the top ball there matches its colour, or the tube is empty — and the tube has not exceeded its capacity.
- ~~The player can undo moves and reset the board; both actions carry a scoring penalty.~~
- Completion is detected automatically when all tubes contain only a single colour from top to bottom.
- Scoring system (capped at 100 pts):
 - Base score: 60 points awarded upon successful completion.
 - Efficiency bonus: up to 25 points based on fewer moves relative to the optimal move threshold.
 - Speed bonus: up to 10 points for faster completion.
- Difficulty scaling: Increase the number of tubes, number of colours, or stack depth in later stages.

Reference: [Ball Sort Puzzle — Yandex Games](#)

Game 3: Block Fill

Inspired by: "Flow Free" (Big Duck Games), the Hamiltonian path problem in graph theory, and puzzle games that require complete path traversal.

Objective: Draw a single continuous path — without lifting your finger — that passes through every "enabled" cell in a 8x8 grid.

- The screen shows a 8x8 grid (64 cells total).
- The grid will contain dots of multiple colors. There will always be 2 dots of the same color.
- The player is expected to draw a line connecting both the dots. A line cannot intersect with another line.
- Resetting the board is allowed.
- Result: All dots connected = Win. Score = Time Window minus Time Taken.
- Puzzle sets are pre-validated to guarantee at least one valid solution exists per configuration.
- Difficulty scaling: Increase the grid size.

Reference: [Flow Mania — CrazyGames](#)

Game 4: MCQ — Mathematics

Inspired by: Competitive math challenge formats such as MATHCOUNTS, Vedic Math quizzes, and adaptive platforms like Khan Academy's mastery system.

Objective: Answer as many mathematics MCQs as possible within 2 minutes, across an adaptive difficulty range.

- Question bank covers: arithmetic, basic algebra, percentages, ratios, geometry, and number theory at the Easy level; progressively introducing speed-maths, data interpretation, and logical number sequences at Medium/Hard.
- Same adaptive difficulty engine as Stage 1 MCQ applies (2 correct → level up; 2 incorrect → level down).
- Questions may include numerical diagrams, graphs, or geometric figures as part of the image-answer system.
- No negative marking. Score = sum of all correct answers in the allotted window.

STAGE 3 | "The Mind Labyrinth" तृतीय

चरण | मन का महाभलभू लयै ा

"The maze has no walls — it is built entirely inside your head."

Stage 3 is the convergence stage — the first to simultaneously test correctness and reaction time. This stage evaluates higher-order thinking: logical deduction, spatial navigation, and associative memory. Inspired by escape room design principles and dual-task cognitive experiments, this is where the cognitive elite begin to separate from the rest.

Stage 3 Game Summary Table

#	Game	Time Window	Evaluation Criteria	Shunya Points
1	MCQ — Logical Reasoning	2 mins (max)	Correct Answers	Easy: 5 Med: 10 Hard: 20
2	Maze Navigation	2 mins (max)	Win/Lose + Reaction Time	Win: 50 pts + (Window – Time Taken)

3	Memory Card Matching	3 mins (max)	Win/Lose + Reaction Time	Win: 50 pts + (Window – Time Taken)
4	MCQ — Mythology	2 mins (time-bound)	Correct Answers	Easy: 5 Med: 10 Hard: 20

Note on Merged Pool: Upon completing Stage 3, all remaining weekly cohorts are merged into a single participant pool. Stage 4 onwards is a unified competition.

Game 1: MCQ — Logical Reasoning

Inspired by: Competitive aptitude formats used in CAT, GMAT, and Mensa IQ tests. Similar to game mechanics in "Brain It On!" and the Lumosity Logic series.

Objective: Solve logical reasoning MCQs that test pattern recognition, deductive reasoning, syllogisms, and analogical thinking.

- Question types include: number series, letter patterns, analogies (A:B :: C:?), odd-one-out, coding-decoding, and logical deduction (If-then statements).
- Same adaptive difficulty engine applies. Questions are strictly text-based or accompanied by diagrams/grids.
- No negative marking. Score = sum of correctly answered questions within 2 minutes.

Game 2: Maze Navigation

Inspired by: Classic maze puzzle games and algorithmic maze-solving challenges (DFS/BFS pathfinding). Cognitive research on spatial navigation and mental rotation tasks.

Objective: Navigate a player token from the entry point to the exit of a procedurally generated maze as quickly as possible.

- The maze is displayed on screen in full at the start of the attempt.
- The player uses swipe gestures (Up, Down, Left, Right).
- Dead ends are incorporated as deliberate traps; the player must backtrack.
- The maze has exactly one optimal solution path. Multiple routes may exist, but only one is shortest.
- Result: Reaching the exit = Win (50 base pts + reaction time bonus). Failure to exit within 2 minutes = Lose (0 pts).

Reference: [Maze: Path of Light — Poki](#)

Game 3: Memory Card Matching

Inspired by: The foundational "Concentration" card game and its digital adaptations. Scientifically linked to episodic memory and working memory research (Baddeley, 1992).

Objective: Find all matching pairs of cards on a face-down grid as quickly as possible, using memory to minimise the number of flips.

- The grid contains an even number of cards (e.g., 4x4 = 16 cards, 8 unique pairs).
- At the start of each round, all cards are briefly revealed face-up for 3 seconds, then flipped face-down.
- The player taps any card to flip it, then taps a second card attempting to find its match:
 - Matched pair: Both cards remain face-up and are locked in place.
 - Non-matching pair: Both cards flip back face-down after a 1-second display.
- Card content types: Symbol pairs, colour pairs, image pairs, or word-image pairs.
- Result: All pairs matched = Win (50 base pts + reaction time bonus). Failure to match all within 3 minutes = Lose (0 pts).
- Difficulty scaling: Increase grid size.

Reference: [Memory Card Game — SolitaireBliss](#)

Game 4: MCQ — Mythology

Inspired by: Cultural knowledge quizzes as seen in "Jeopardy!" category design and specialised mythology trivia apps (e.g., "MythMaster").

Objective: Answer mythology-themed MCQs spanning Hindu, Greek, Roman, Norse, Egyptian, and global folklore traditions.

- Question types: Identify a deity from a description, match a mythological weapon to its owner, complete a legend, or identify a symbol or artefact.
 - Questions and answer options may include illustrative artwork or iconography.
 - Same adaptive difficulty engine applies. Score = sum of correctly answered questions within 2 minutes.
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STAGE 4 | "The Crucible of Will"

चतुर्थ चरण | संकल्प की अि न

"Skill brought you here. Only those who refuse to break will leave."

Stage 4 marks the beginning of the post-bucketing phase, where all surviving cohorts compete together in a unified pool for the first time. This stage introduces a new dimension: persistence. Games here have no hard win conditions — players are tested on how long they can sustain performance, how deep they can go, and how composed they remain under escalating pressure.

Stage 4 Game Summary Table

#	Game	Time Window	Evaluation Criteria	Shunya Points
1	MCQ Rapid Fire (2 incorrect = out)	5 mins (max)	Correct Answers	10 pts per correct answer
2	Skill Memory Test — Sequence (3×3)	5 mins (max)	Levels Cleared + Speed	2 × Levels + (Level × 2s per attempt)
3	Mini Sudoku (Easy)	5 mins (max)	Win/Lose + Reaction	Win: 100 pts + (Window – Time Taken)
4	Path Optimisation	5 mins (max)	Win/Lose + Efficiency	Win: 100 pts + efficiency bonus

Game 1: MCQ Rapid Fire (2 Incorrect = Out)

Inspired by: "Who Wants to Be a Millionaire" elimination format and "Fastest Finger First" rapid-answer game mechanics.

Objective: Answer as many MCQs as possible within 5 minutes — but two cumulative incorrect answers will immediately end the session.

- Questions are delivered in rapid succession with minimal pause between them.
- Theme: Mixed — covering all prior MCQ themes plus current affairs, science, and technology.
- A persistent "Strike Counter" is visible on screen (0/2 incorrect). Two strikes at any point terminate the player's session immediately.

- Score = 10 pts per correct answer. No partial credit. No adaptive difficulty — questions are served at a fixed randomised difficulty.
- This game directly rewards breadth of knowledge and sustained performance under psychological pressure.

Game 2: Skill Memory Test — Sequence (3×3)

Inspired by: "Simon Says" electronic games and the N-Back working memory paradigm widely used in cognitive psychology research.

Objective: Observe and reproduce an increasingly complex sequence of highlighted cells on a 3×3 grid, advancing through levels until a mistake is made.

- Level 1: A sequence of 3 cells lights up in order. The player must tap them back in the same order.
- Each subsequent level adds one additional cell to the sequence.
- Scoring: $2 \times (\text{Number of Levels Completed})$. Each level also has a time component — $\text{Level} \times 2$ seconds per attempt window. Faster completion earns bonus time credit.
- A single mistake on any level ends the game immediately. No second chances within the session.
- Maximum sequence length can reach 12–15 cells at the highest levels.

Reference: <https://humanbenchmark.com/tests/sequence>

Game 3: Mini Sudoku (Easy — 6×6)

Inspired by: Classic Sudoku (popularised globally post-2004). Digital implementations include NYT Sudoku, Sudoku.com, and competitive Sudoku apps with sub-60s leaderboards.

Objective: Complete a 6×6 Sudoku grid as quickly as possible.

- A 6×6 grid is pre-populated with a valid, uniquely solvable puzzle (typically 12–15 starting clues for Easy difficulty).
- Each row and each column must contain the digits 1–6 exactly once.
- The grid is divided into six 2×3 sub-boxes, each also requiring the digits 1–6 exactly once.
- Input method: Tap a cell → tap a digit from the 1–6 keypad. Pencil-marking (candidate notes) is available as a helper tool.
- Result: Correctly filled grid = Win (100 pts + reaction time bonus). Incorrect submission or timeout = Lose (0 pts).
- Progression note: Stage 5 introduces standard 9×9 Sudoku at Medium or Hard difficulty.

Game 4: Path Optimisation

Inspired by: The Travelling Salesman Problem (TSP), "Cut the Rope" routing mechanics, and IBM's annual "Ponder This" competitive puzzle challenges.

Objective: Find the shortest or most efficient path that visits all required checkpoint nodes on a grid-based network within the time limit.

- The screen displays a grid with a set of checkpoint nodes (5–10 nodes at Easy difficulty) connected by weighted or unweighted edges.
- The player starts at a defined origin node and must visit all checkpoint nodes — either returning to origin (closed-loop variant) or reaching a defined destination (open-path variant).
- The player draws the path by tapping nodes in sequence. The current path length (in steps or cost units) is displayed in real time.
- Multiple valid paths exist; the game rewards the most optimal path found within the time window.
- Result: All required nodes visited = Win. Bonus points are awarded for path efficiency relative to the pre-calculated optimal solution.
- Difficulty scaling: Increase node count, introduce weighted edges, or add blocked and dynamic edges.

STAGE 5 | "The Final Reckoning"

पंचम चरण | अंतिम परीक्षा

"Everything you learned, everything you endured — this is where it counts."

Stage 5 is the final online stage and the last round of the post-bucketing phase. It does not introduce entirely new game types — instead, it re-examines the same core skills tested across Stages 1–4, but at elevated difficulty and with tighter time pressure. Players who reach Stage 5 have already proven speed, memory, and knowledge. Now they must prove mastery. Only a few hundred will survive to receive their invitation to the Offline Finale.

Stage 5 Game Summary Table

#	Game	Time Window	Evaluation Criteria	Shunya Points
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1	Connections	5 mins (max)	Correct Groups + Streak	5–10 pts per group; streak bonuses apply
2	Sliding Puzzle (Advanced) (3x3 without numbered tiles)	5 mins (max)	Win/Lose + Reaction Time	Time Window – Time Taken
3	Maze Navigation (Hard)	5 mins (max)	Win/Lose + Reaction Time	Win: 50 pts + (Window – Time Taken)
4	MCQ Rapid Fire (2 incorrect = out)	5 mins (max)	Correct Answers	10 pts per correct answer

Game 1: Connections

Inspired by: The New York Times "Connections" puzzle — a category-grouping game that tests vocabulary, lateral thinking, and cultural knowledge simultaneously.

Objective: On a 3×3 grid of tiles, continuously identify and connect groups of related items (by theme, category, pattern, or logic) within a fixed 2-minute session window.

- **Game Loop:** Players identify valid groups of tiles that share a hidden connection (e.g., types of birds, words that follow "fire," things associated with a particular deity). Correct groups clear from the board and new tiles dynamically replace them.
- **Time Constraint:** Each session lasts approximately 2 minutes, rewarding rapid pattern recognition and quick decision-making.
- **Scoring:** Each correct connection awards 5–10 points. Consecutive correct groups trigger a streak bonus, enabling a skilled player to accumulate approximately 100 points in a full session.
- **Replayability:** Infinite tile regeneration with increasing ambiguity and difficulty ensures there is no hard score cap — players continue scoring until time expires.
- **Difficulty scaling:** Increase the subtlety of the category connections (e.g., words with double meanings, cross-cultural references) in harder rounds.

Reference: [Connections — New York Times Games](#)

Game 2: Sliding Puzzle (Advanced)

Inspired by: The classic 15-Puzzle. Stage 5 variant increases difficulty with a larger grid and reduced reference time.

Refer to the explanation provided above.

Game 3: Maze Navigation (Hard)

Inspired by: Algorithmic maze generation (recursive backtracking, Prim's algorithm). Hard variant introduces fog of war and larger grid dimensions.

Objective: Navigate a player token from entry to exit.

- Increased grid size.
- Controls remain the same: swipe gestures
- Dead ends are more frequent and deliberate in the hard layout. Backtracking is expected and factored into the time window.
- Result: Reaching the exit = Win (50 base pts + reaction time bonus). Failure within 5 minutes = Lose (0 pts).

Reference: [Maze: Path of Light — Poki](#)

Game 4: MCQ Rapid Fire (2 Incorrect = Out)

Inspired by: "Who Wants to Be a Millionaire" elimination format and "Fastest Finger First" rapid-answer game mechanics.

Objective: Answer as many MCQs as possible within 5 minutes — but two cumulative incorrect answers will immediately end the session.

- Same rules as Stage 4's Rapid Fire, but drawn from a harder question bank skewed towards Medium and Hard difficulty tiers.
- Theme: Mixed — all prior topics. Questions may deliberately combine themes (e.g., a logical reasoning problem framed in a mythology context).
- Strike Counter visible on screen (0/2). Two strikes = immediate session end.
- Score = 10 pts per correct answer. No partial credit. No adaptive difficulty.

Appendix: Cross-Stage Mechanics

Scoring Philosophy

Shunya uses a composite scoring approach that rewards distinct cognitive dimensions depending on the stage:

Stage	Primary Dimension	Secondary Dimension	Tertiary Dimension
Stage 1	Correctness (memory accuracy)	Knowledge breadth	—
Stage 2	Reaction time	Spatial problem-solving	—
Stage 3	Correctness + Speed (dual)	Spatial navigation	Cultural knowledge
Stage 4	Persistence under pressure	Strategic depth	Time optimisation
Stage 5	Mastery synthesis	Speed under familiarity	Pattern recognition

Adaptive MCQ Engine — Detailed Behaviour

Condition	Action	Points per Question
Starting state	Begin at Easy	5 pts
2 consecutive correct answers	Level increases (Easy → Med → Hard)	10 pts / 20 pts
Condition	Action	Points per Question
2 consecutive incorrect answers	Level decreases (Hard → Med → Easy)	10 pts / 5 pts
Already at min/max level	Level stays; streak counter resets	Unchanged

Game Integrity & Anti-Cheat Considerations

- All puzzle configurations (mazes, sliding puzzles, Sudoku, Block Fill) are server-side validated to guarantee exactly one solvable state per session. (For games with efficiency marks, all possible solutions should be validated)
- Reaction time scoring implements a 0–5 second grace-period floor to eliminate exploit scenarios (e.g., pre-clicking, auto-bots).

- MCQ question banks must be rotated per cohort-session to prevent question farming or sharing across users.
- Session tokens should be invalidated immediately upon detecting abnormal input timing patterns (sub-100ms consistent responses across 10+ MCQs).

Guide

- To make users familiar with games it's expected that before the game is started a 5-10 second animation (not video) is displayed with a sample game step.

Game Reference Links

The following external games and frameworks were referenced during game design for Shunya:

- Simon Says (Hasbro, 1978) — Sequence memory benchmark for Sequence Recall design.
- Concentration / Memory — Classic card-matching reference for Memory Card Matching.
- Ball Sort Puzzle (AppStore, 2020+) — Reference for Colour-Based Sorting mechanics.
- Flow Free (Big Duck Games) — Reference for Block Fill path-drawing mechanics.
- Sudoku.com / NYT Sudoku — Reference UX implementations for Mini Sudoku.
- Corsi Block-Tapping Test (Corsi, 1972) — Cognitive psychology basis for Sequence Recall.
- Kahoot! / Quizlet Live — Adaptive quiz scoring inspiration for the MCQ engine.
- IBM Ponder This (ibm.com/community/ponder) — Path optimisation puzzle design reference.
- NYT Connections (nytimes.com/games/connections) — Category grouping design for Connections game.
- CrazyGames "What's the Difference" — Visual discrimination reference for Spot the Difference.